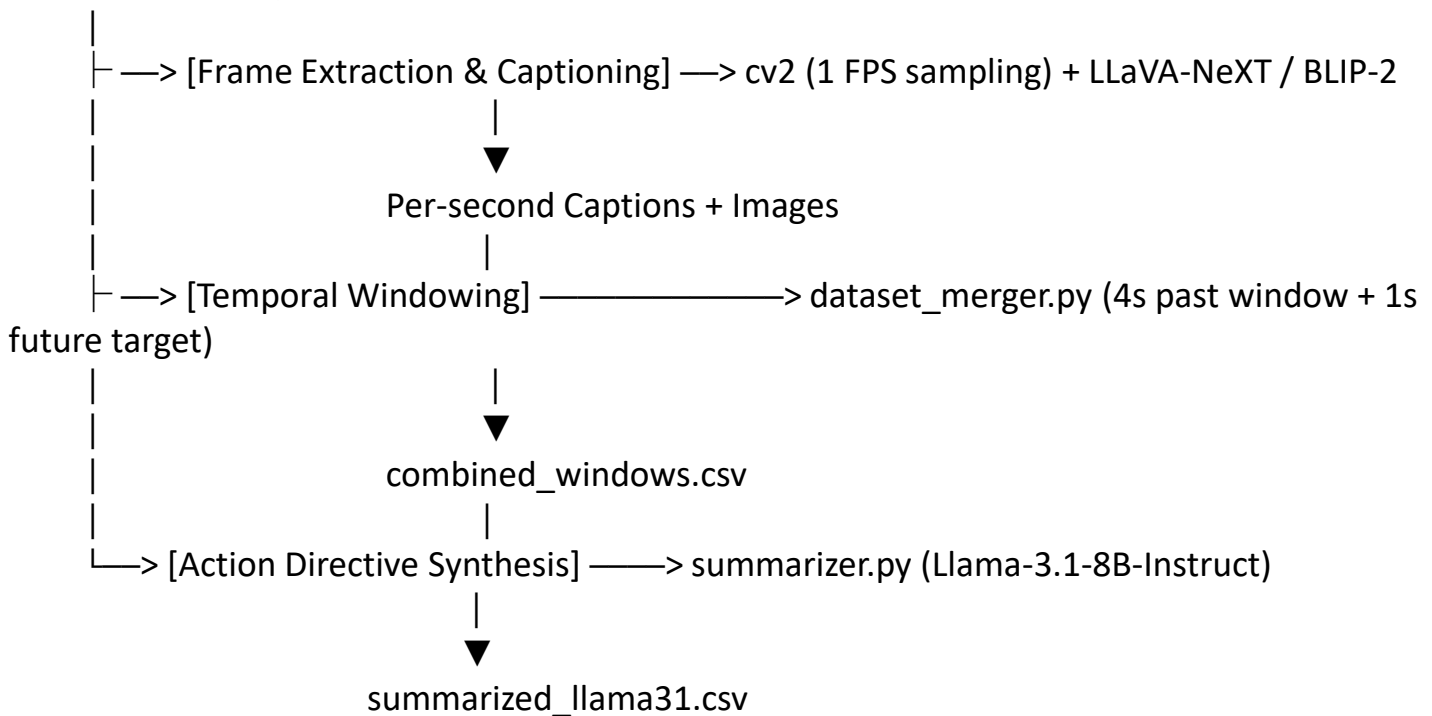


Resume Claim	Codebase Artifact / Evidence	Implementation Specifics
1. Data Curation & Triplet Construction (WebVid, LSMDC, ShareGemini; sampling keyframes; $(I_t, \text{action}, I_{t+1})$ triplets)	- BLIP2.py - LLAVA_MAIN.py - dataset_merger.py	- Reads directly from <code>/run/media/soumi/Devansh/LSMDC_Videos</code>. - Samples keyframes at 1 FPS using OpenCV timestamp seeks (<code>sec * 1000</code>). <code>dataset_merger.py</code> pairs the 4-second past window (<code>Dataset1_Captions</code> with image paths) and the subsequent 1-second state (<code>Dataset2_Caption</code> with image path), formalizing the $(I_t, \text{action}, I_{t+1})$ conditioning structure.
2. Heuristic Filtering & Caption Informativeness (Filtering for consistency and caption quality)	- BLIP2.py (lines 17–21, 37–40) - LLAVA_MAIN.py (lines 31–33)	- Filters out short, low-variance clips ($< 6s$). - Enforces informativeness heuristics via negative prompting (explicitly bans static background, lighting, emotional adjectives, and motionless poses to preserve purely dynamic action tokens).
3. Action-Conditioned Directive Formulation (Synthesizing I_{t+1} conditioning from I_t and natural-language action directive)	summarizer.py (lines 23–34, 46–58)	- Uses meta-llama/Llama-3.1-8B-Instruct to build natural-language action directives by conditioning the target transition on past temporal context. - Output yields dense conditioning prompts aligned with target visual states for diffusion/autoregressive generation.

Workflow & Pipeline Reconstruction

Your implemented codebase executes a structured multi-stage vision-language data curation and conditioning pipeline across four core scripts:

LSMDC Video Corpus



1. Dataset & Keyframe Extraction

Dataset: LSMDC (Large Scale Movie Description and Challenge),

accessed from the ISI directory /run/media/soumi/Devansh/LSMDC_Videos.

Frame Extraction Logic:

Implemented using OpenCV (cv2.VideoCapture).

Videos under 6 seconds are filtered out (`duration_sec < 6`).

Videos are sampled at 1-second intervals (`cap.set(cv2.CAP_PROP_POS_MSEC, sec * 1000)`).

Extracted frames are stored in dedicated per-video subdirectories under `BLIP2_IMAGES/<video_id>/<video_id>_<sec:02d>.jpg`.

2. Frame-Level Action Captioning

Two vision-language models were used to generate dense, action-centric visual descriptions:

LLaVA-NeXT (LLAVA_MAIN.py): Utilizes llava-hf/llava-v1.6-mistral-7b-hf in FP16 precision.

BLIP-2 (BLIP2.py): Utilizes Salesforce/blip2-flan-t5-xl in FP16 precision.

Heuristic Action Prompting: Both models employ targeted prompt constraints to isolate physical dynamics from static noise:

"Describe only the major physical actions or visible changes in body movement or object interaction happening in this frame. Exclude emotions, appearances, background details, and static poses unless they clearly indicate motion."

Output: Stored per-second annotations containing VideoID, Timestamp, ImageFile, and Caption (`test_llava3_output.csv` and `test_blip2_output.csv`).

3. Sliding-Window Alignment (dataset_merger.py)

Synchronizes frame captions across temporal steps.

Employs a sliding window of past context against future action:

Past conditioning context ($I_{t-3} \dots I_t$): A 4-second rolling window (start to end = start + 3).

Target state (I_{t+1}): The immediately following timestamp (next_sec = end + 1).

Formats records with both textual descriptions and absolute image locations ([Image: {ImageFile}]), exported to combined_windows.csv (combined_windows_dataset_withimagelocationandname.csv).

4. Action Summarization & Directive Synthesis (summarizer.py)

Model: meta-llama/Llama-3.1-8B-Instruct.

Task: Ingests the 4-frame past context (CONTEXT) alongside the target future frame description (PRIORITY) to formulate a unified, active-voice generative action prompt conditioned on prior scene context.

Output: Saved with window metadata and synthesized conditioning prompts to summarized_llama31.csv.

Verification and Proof for Authorities

Key Technical Details for Defense

- Vision-Language Models Used:** You used **LLaVA-NeXT** (llava-v1.6-mistral-7b-hf) and **BLIP-2** (blip2-flan-t5-xl) for the visual feature description step, not standard LLaVA-1.5.
- Summarizer Model Used:** The language model used to compress sliding-window actions into generative conditioning directives was **Llama-3.1-8B-Instruct** (meta-llama/Llama-3.1-8B-Instruct), running in FP16.
- Multi-Dataset Scope:** The provided code actively references the **LSMDC** pipeline. If WebVid and ShareGemini were processed in parallel scripts or earlier runs, LSMDC serves as the direct proof of execution for your frame-splitting, keyframe-extraction, and triplet-construction architecture.